scientific reports

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Attention-based hybrid contrastive learning for unsupervised person re-identification

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Existing unsupervised Re-ID methods often rely on pseudo-labels generated by clustering algorithms. However, the effectiveness of these methods is highly dependent on the quality of the learned features. As a result, improving the quality of the extracted features will highly enhance the performance of the model. In this paper, an attention-based hybrid contrastive learning method (AHCL) is proposed by incorporating a novel attention mechanism, which combines spatial attention and channel attention to enable the model to learn more effective features. Additionally, a hybrid contrastive learning approach is introduced, which utilizes a memory dictionary to store features and extracts both cluster-level and instance-level features for contrastive learning. For cluster-level contrastive loss, it ensures that the network is updated in a more stable way by comparing the sample with the cluster center. Meanwhile, it can extract more discriminative information by comparing the sample with hard samples in instance-level contrastive loss. Thus, the performance of the model would be improved by using the combination of this two loss. The experimental results on three widely-used large-scale Re-ID datasets and experimental settings demonstrate that the proposed AHCL outperforms the compared existing methods and significantly enhances the accuracy of unsupervised person Re-identification.

Unsupervised person Re-identification (Re-ID) is a challenging task that involves matching pedestrian images from various camera viewpoints without the need for labeled data. This task has acquired significant attention in recent years due to its applications in real world. The most significant challenge in unsupervised Re-ID is dealing with large variations in viewpoint, lighting conditions, weather changes, cluttered backgrounds, and other factors without the aid of supervised labels.

Currently, the existing unsupervised person Re-ID methods are usually proposed based on weakly supervised information. For MMT[1], pseudo-labels generated by DBSCAN[2] algorithm are used to train mutual mean teaching networks. In[3], hierarchical clustering algorithm is used to assign pseudo-labels to the training samples for HCT. In[4], the k-means clustering algorithm is used when SSG deals with multi-view training samples. However, the quality of the pseudo-labels generated by clustering algorithm largely determines the performance of the model, while the training samples directly affects the quality of the pseudo-labels. Most existing unsupervised person Re-ID methods use a fixed feature extractor to extract pedestrian features, which often lack robustness and fail to consider the importance of attention mechanisms. In order to selectively enhance the effective regions of the image and ignore other less informative regions, the attention mechanism could be incorporated into the feature extraction process. Thus, the more robust and stable features would be extracted, which can improve the performance in unsupervised person Re-ID tasks.

Recently, methods based on contrast learning have been widely used in the unsupervised feature learning process. In these methods, a self-supervised mechanism based on data augmentation is used to learn the representative information in features. In the same batch of SIMCLR[5] method, each sample and its samples generated by data augmentation are considered as positive samples while the others are defined as negative samples. Besides, the contrastive learning is used to discriminate the positive and negative samples to prevent the model from reaching a trivial solutions. In order to train the network more efficiently, the more negative samples need to be accommodated in SIMCLR and the size of batch are set from 256 to 4096. Byol[6] and Simsiam[7] do not use negative samples compared to traditional comparison learning methods, but instead use a prediction layer to prevent feature collapse. In SWV[8], clustering is used to prevent feature collapse. Compared with the traditional comparison learning methods, a prediction layer is used to prevent feature collapse in Byol[6] and Simsiam[7]. The

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improvement has been achieved on ImageNet[9], but the traditional contrast learning methods are not suitable for unsupervised person recognition because of the severe feature collapse phenomenon.

In order to address the problems with existing methods, a novel and effective framework, called attention-based hybrid contrastive learning model (AHCL), is proposed to extract more robust feature information for the performance improvement of clustering in this paper. In AHCL, the enhanced feature extractor can be used to extract more robust features, which are stored by using memory bank. Meanwhile, the hybrid contrastive learning is used to extract cluster-level features and instance-level features from the memory bank to improve the performance. The distance between the sample and the centre of this cluster can be reduced by comparing the selected sample with the centre of the cluster in which it is located, making the samples in the same cluster more compact. By comparing samples from different clusters, the distance between confusable samples can be increased. Thus, the misclassification rate of the model would be reduced.

It is worth noting that the effectiveness of the features greatly determines the performance of the Re-ID model. For the existing methods, the pedestrian feature is always extracted by a fixed feature extractor, which may be the lack of robustness. In this paper,the attention mechanism is proposed to extract the feature, and then an efficient feature processing model is used to improve the performance. The channel attention-based pooling feature fusion model (CAPF) is used to enhance the capability of useful information extraction for the feature extractor. The feature fusion part of the AHCL network is constructed by combining the pooling fusion strategy (GAP + GMP) with the channel attention model. Extensive experiments on three benchmark datasets demonstrate that our proposed approach is effective. Our contributions in this paper can be summarised as:

1. We propose a novel structure, in which the attention mechanism and different pooling methods are integrated. Compared to the traditional feature extractors, the proposed structure dramatically improves the robustness of the features.
2. A novel contrastive learning framework is proposed to solve the problem of unsupervised person re-identification. This framework simultaneously utilizes cluster-level features and instance-level features for contrastive learning.
3. Experiments on different databases have demonstrated that the proposed method achieves significant improvements for unsupervised person Re-identification.

Related work

Unsupervised person re-identification

Unsupervised person Re-identification methods typically consist of three stages: generation of pseudo-labels by clustering algorithm, initialization of the memory dictionary, and training of the neural network. The above stages are extensively used in most of the effective unsupervised methods[10]. At the beginning of each epoch, the features of the training data are extracted by the current neural network and stored in memory dictionaries. The image features are then clustered and pseudo-labels are generated by using clustering algorithms, such as DBSCAN[2] or K-means[11]. Each image is also assigned a cluster ID as its identity simultaneously. Finally, the neural network is trained by using memory dictionary-based contrastive losses, e.g., triple loss[12], InfoNCE loss[13], or other nonparametric classification losses[14].

The better results on the specific datasets have been achieved by using the unsupervised methods. In[15], BUC considers each instance as a separate category, and then progressively merges the different clusters. In order to reduce the effect of pseudo label noise, The similarity-based soft labels are used by SoftSim[16]. JVTC[17] considers each sample as a cluster. The similar samples are treated as a cluster and assigned the same pseudo-labels. In order to solve the multi-label classification problem, the memory-based nonparametric multi-label classification loss method (MMCL)[18] is proposed to predict the quality of pseudo-labels, including similarity computation and cyclic consistency. For MMT[1], a novel softmax-triplet loss is proposed to use the soft pseudo-triplet labels for reducing the effect of noise in pseudo labels. RDEIC-LFW-DSS[19] provides a robust framework for representation weighting and neighbor-based clustering. EDCWRN[20]introduces local feature weighting and dynamic sample selection, offering valuable insights into feature-weight learning. SSFCM-FWCW[21]highlights the importance of feature and cluster weighting in semi-supervised settings.The above methods demonstrate that the robustness features can enhance the performance of the pseudo-label-based methods for the unsupervised person re-identification.

Memory dictionary

For the approach based on the contrastive learning[22], the encoder is trained for searching a desired object in a memory dictionary. For unsupervised visual representation learning[23–25], some research efforts have also shown that it is beneficial to construct dynamic dictionaries, which are continually updated to facilitate contrastive unsupervised learning. Similarly, the contrastive learning[17,26] approach based on memory dictionary is always used by the state-of-the-art unsupervised person re-identification methods. In SPCL[26], a novel self-paced contrastive learning framework is proposed to refine memory dictionary features by creating clusters with higher reliability. In order to enhance the contrastive learning, the pairwise similarity between instances is used in ICE[27]. For PPLR[28] , a cross-consistency score is used to calculate the similarity between global and local features, which can refine the pseudo-labels. In Cluster-contrast[29], the features are stored, and then a cluster-level Contrast loss is computed to update each cluster consistently. In the training process, query instance features could dynamically update the instance feature vectors in memory dictionary.

Attention mechanism

In the person re-recognition model, the attention mechanism is used to emphasize representative informative features of the image for solving some problems, such as the differences in pedestrian, occlusion, incomplete

bounding box[30–32], and so on. There are two attentional mechanisms, which are often used in person re-identification task, e.g., channel attention and spatial attention. The spatial attention allows models to selectively focus on more important features in the spatial dimension, while the channel attention improves characterization by making the model focus on more important channels. In some researches, these two mechanisms are combined. For example, the attention regions and feature representations are jointly learned by combining spatial information in the samples and channel information at different scales[33]. It is worth noting that these attention mechanisms are usually used in the supervised learning, whereas they are used for the unsupervised learning in this paper, which will be a challenge task.

Proposed method

Traditional attention-based methods usually rely on labels that guide the neural network toward the discriminative features in an image. However, in unsupervised learning, label information needs to be improved, and there is a need for attention modules that can effectively locate and embed relevant features. To enhance the robustness of feature extraction and improve the quality of features, we propose a channel attention-based pooling feature fusion model (CAPF). Moreover, we add it to the ResNet50[34]-based feature extractor, which is commonly used in traditional unsupervised person re-identification methods.

After extracting the enhanced features, we use the DBSCAN[2] algorithm to cluster the extracted person features. The clustering algorithm groups similar features and assigns them the same pseudo-label. Subsequently, the loss function of the network is computed based on the pseudo-label and is used to optimize the discriminative power of the neural network. During the network training process, we use the method in[25] and a memory dictionary to store the feature vectors. Sample center features of the clustering are calculated from the dictionary to update the storage of sample centers. Also, complex clustering samples are extracted for contrastive learning, and the dictionary is updated after training. Finally, we use the modified InfoNCE loss[35] to update the dictionary. We describe the details of this method in Section . The InfoNCE loss is expressed as follows:

$$L_q = -log\frac{exp(q \cdot k_+/\tau)}{\sum_{i=0}^{N} exp(q \cdot k_i/\tau)} \tag{1}$$

where q is an encoded query selected from the gallery k_1, k_2...k_N , and k_+ is a positive feature with the same label as q. The temperature parameter τ,which is empirically set as 0.05(consistent with SPCL[26]), controls the distance between different clusters. We illustrate the whole training strategy proposed in the paper as Fig. 1.

Attention-based feature augmentation

We first introduce the attention-based feature enhancement mechanism. Utilizing useful information in neural networks is a vital issue to be solved in person re-identification research. We proposed the channel attention-

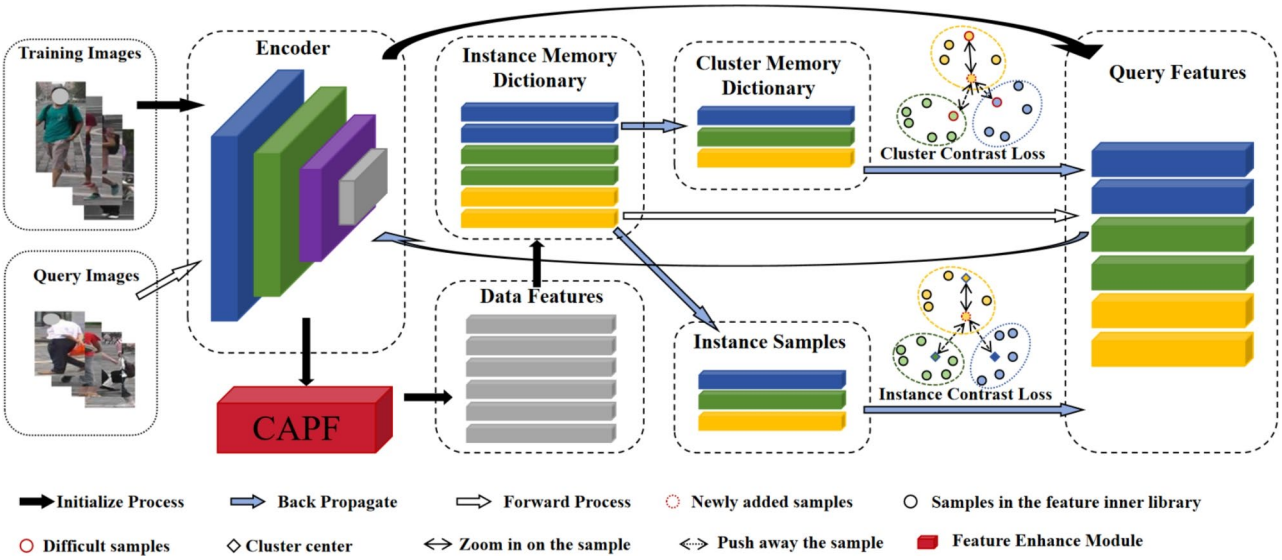


Fig. 1. In the training process of the proposed method, the features are first extracted from the training image by Encoder, and then the feature map is weighted and fused by the CAPF module to enhance the features, and the clustering algorithm is used to cluster them and store them in the Instance memory dictionary. Secondly, the instance memory dictionary and cluster memory dictionary were initialized, and the cluster memory dictionary and instance samples were extracted according to the features in the instance memory dictionary for comparative learning. Thirdly, the features of the queried images in each batch were extracted, and then the cluster contrast loss and instance contrast loss were calculated separately. Finally, update the encoder and the two memory dictionaries.

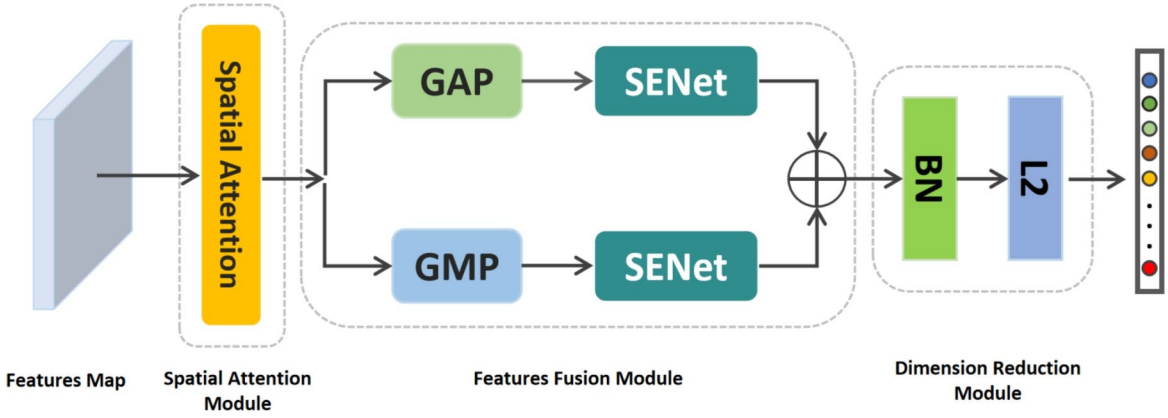


Fig. 2. The channel attention-based pooling feature fusion model (CAPF) comprises of a spatial attention module, a feature fusion module which combining the pooling fusion strategy (GAP + GMP) with the channel attention model, and a dimension reduction module. This model is designed to extract more robust features from the feature maps.

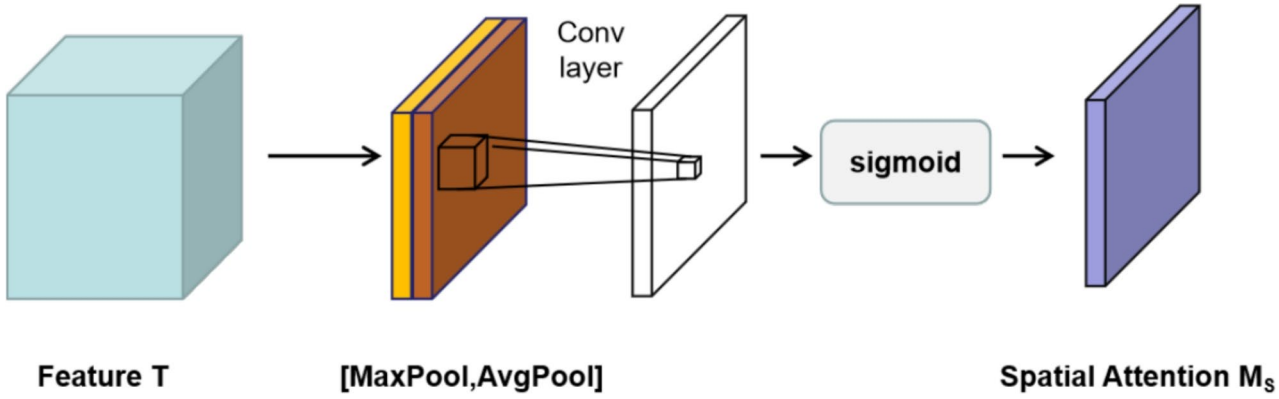


Fig. 3. The structure of the spatial attention module.

based pooling feature fusion module (CAPF) as a feature enhancement module to address this issue. We apply this module to the last layer of ResNet50 to enhance the robustness of extracted pedestrian features.

The CAPF module consists of three parts: spatial attention module, feature fusion module, and dimension reduction module. Figure 2 illustrates the structure of this module. When feature maps enter the CAPF module, they first go through the spatial attention module to remove the background noise and then pool it using different pooling methods (GAP and GMP). Subsequently, the channel attention module enhances the features, fusing the features from the two pooling methods.

Spatial attention module

The feature map T extracted from ResNet50 first passes through the spatial attention module, which can give higher weight to the pedestrian while reducing the weight of other regions. This reduces the interference of background noise information. Figure 3 shows the structure of the spatial attention module, assuming that the input feature map T has channel C, row height H, and column width W. We select a channel to use average pooling and maximum pooling respectively, generate two feature vectors with dimensions of $H \times W \times 1$. This option is equivalent to reducing the number of the feature map channels T to 1. Next, these two feature vectors are spliced to form a new feature vector M, and a 7×7 convolutional layer is added behind M. The sigmoid activation function computes the weighted feature map M_s. Finally, the weighted map M_s is multiplied with the feature map T to obtain the new feature after spatial attention weighting:

$$M_S(T) = \sigma\left(f^{7\times7}\left([AvgPool(F), MaxPool(F)]\right)\right) = \sigma\left(f^{7\times7}\left(\left[F^S_{avg}; F^S_{max}\right]\right)\right) \quad (2)$$

where σ is the sigmoid operation and 7×7 is the size of the convolution kernel.

Features fusion module

Among the basic research methods for person re-identification, Global Average Pooling (GAP) and Global Max Pooling (GMP) are widely used as feature pooling methods. Global average pooling can enhance the relationship

between the extracted features and the original person labels while retaining the background information. Global max pooling selects apparent parts of features and retains the most distinguishable feature in the image. Current methods usually use only one of these methods but fail to combine the advantages of both methods. This leads to the loss of part of the information in the feature map, so the representation of the extracted features is poor. Therefore, adding them up preserves the salient features of the image without the loss of global information. This makes the feature vectors extracted from the feature map more representative and comprehensive, and further enhance the performance of the person re-identification network.

After each pooling operation, we use SENet for feature enhancement; the specific structure of SENet[36] is shown in Fig. 4. Essentially, SENet is implemented by doing an attention operation on the channel dimensions. It allows the model to focus more on channel features with more significant information content and reduces the influence of irrelevant channel features. The main steps of SENet consist of two parts: compression and excitation. However, because of the limited range of the convolution operation, it is difficult to obtain enough information to establish the information correlation between channels, and its sense field is relatively narrow. To address this issue, compression transforms all spatial features on each channel into a single $1 \times 1 \times C$ global feature and generates channel statistics using global average pooling. After that, a sigmoid activation function is used to capture the correlation between channels. We then fused the features generated from the two paths to obtain a fused feature vector. The fused feature vector not only highlights the representative information but also preserves the background and texture information, thus increasing the amount of feature information.

Dimension reduction module

The dimension reduction module is located at the end of the feature enhancement module. It aims to reduce the dimensionality of the fused feature vector to a fixed size of 2048 dimensions. The fused feature vector passes through a fully connected layer, followed by a batch normalization[37] layer and a L2 normalization layer. The fully connected layer maps the high-dimensional fusion feature vectors to a low-dimensional space, helping to reduce overfitting and improving the generalization ability of the model simultaneously. The batch normalization layer reduces internal covariate bias in deep neural networks during training, improving model robustness. Finally, the L2 normalization layer converts feature vectors into unit lengths, which makes computing distances between them simpler during model training process. Using the dimension reduction module speeds up feature processing, making it more suitable for real-world applications.

Hybrid contrast learning

We first introduce the notations used in this section, given the unlabeled training set X =$x_1, x_2, \ldots, x_N$ with N instances. The corresponding features extracted from the backbone encoder f_θ are denoted by U = $u_1, u_2, \ldots, u_n$ where $u_i = f_\theta(x_i)$. In addition, we use x_i^q to refer to a query instance and x_i^g to refer to a gallery instance. The gallery set x_i^g contains all the retrieved images in the database. We use the query image x_i^q to search the gallery set and use the Euclidean distance$d(x^q, x^g) = \|f_\theta(x^q) - f_\theta(x^g)\|$, to measure the similarity between them, where smaller distances imply higher similarity between the images.

Traditional contrastive learning method usually compares and minimizes the distance between samples in the same category and uses InfoNCE loss as the loss function to update the model. However, these methods merely focus on the relationship between individual instances and do not take into account the link between clusters. To overcome this limitation, we propose a hybrid contrastive learning method that combines cluster-level features and instance-level features. The loss function is shown below:

$$L_{HCL} = (1 - a)L_{ccl} + aL_{icl} \tag{3}$$

where L_{HCL} is the hybrid contrast loss, L_{ccl} is the cluster contrast loss, L_{icl} is the instance contrast loss, and a is the weighting coefficients that balance the contributions of the cluster contrast loss and instance contrast loss. The training details are presented in Algorithm 1.

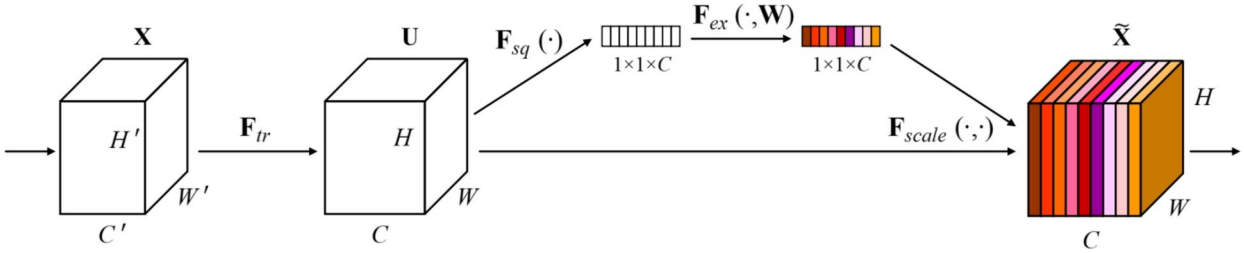


Fig. 4. The structure of the squeeze and excitation networks.

Require: Unlabeled training data X
Require: Initialize the backbone encoder f_θ with ImageNet-pretrained ResNet-50
for n in [1, epochs] **do**
  Extract feature vectors U from X by f_θ, and use CAPF to enhance U
  Clustering U into N clusters with DBScan
  Initialize memory dictionary M_i and M_c
  for n in [1, batchsize] **do**
    Sample $P \times K$ query images from X
    Extract query features and enhance them
    Compute L_{ccl} with Eq. 4 and L_{icl} with Eq. 6, and add them up as L_{hcl}
    Update the memory dictionary M_i and M_c with Eq. 5 and Eq. 7
    Backward to update model f_θ
  end for
end for

Algorithm 1. Attention-based hybrid contrastive learning method for unsupervised re-identification

Cluster level contrastive learning
The traditional contrastive learning method uses a memory dictionary to store instance-level features. We use a similar method, first we extract the features of all instances in each cluster. Then, we compute their averages to obtain cluster-level features. These cluster-level features are represented by cluster centroidsc_1, c_2, ..., c_N. We then use a cluster-level memory dictionary M_c to store them. The cluster-level contrastive loss function we employ is as follows:

$$L_{ccl} = -log\frac{exp(q \cdot c_+/\tau)}{\sum_{i=0}^{N} exp(q \cdot c_i/\tau)} \tag{4}$$

where q refers to the feature of the selected query instance, c_1, c_2, ..., c_N denotes the cluster centroids, c_+ represents the positive cluster currently used, N represents the number of clusters. At the instance level, the memory dictionary is updated as follows:

$$c_i \leftarrow mc_i + (1-m)\tilde{c}_i \tag{5}$$

where m is the momentum updating factor and $\tilde{c}_i$ is the average of the samples in the i-th cluster. The memory dictionary stores all cluster feature vectors and the multi-class score is computed by comparing the query feature q with all cluster feature vectors in the memory dictionary. At the end of each epoch of network training, the memory dictionary is updated.

Instance level contrastive learning
Traditional instance-level contrastive learning method only considers instances in one cluster. In contrast, we extract the most representative samples from each sample for contrastive learning. For samples in the cluster of the query instance, we select the farthest instance from the cluster as the positive sample, thereby enhancing the compactness of samples in one cluster. For samples from a different cluster, we select the closest instance to the query instance among them as the negative sample, thus enhancing the independence between classes. During training, we randomly select P person identities from the training set and take Z instances from each person. We use an instance-level memory dictionary M_i to store all the features and select the appropriate features from them. The cluster-level contrastive loss function we employ is as follows:

$$L_{icl} = -log\frac{exp(q \cdot u_+/\tau)}{\sum_{i=0}^{P} exp(q \cdot u_i/\tau)} \tag{6}$$

where q is the feature of the selected query instance, u_+ is the farthest sample in the same cluster, u_i is the nearest sample among different samples, and τ is the temperature hyper-parameter. The instance-level memory dictionary is updated in the following way:

$$m_i \leftarrow u_i \tag{7}$$

During each update, the instance-level memory dictionary is updated with all the instances in the batch.

Experiments

Datasets

We evaluate the proposed method on three widely used real-world person re-identification (re-ID) benchmarks: Market-1501[38], DukeMTMC-reID[39] and MSMT17[40], all of them are large-scale datasets for person re-identification.

Market-1501 dataset was launched by CUHK in 2015 for the person re-recognition problem, which contains person images covering more than 15,000 people with different identities in a total of 3.9 million images taken by different cameras from shopping malls, supermarkets, schools, and other public places. Each image carries information such as camera ID, person identity tag, and shooting time, which can be used to train and evaluate the performance of person re-recognition algorithms. Due to its data size and diversity, the Market-1501 dataset has quickly become one of the benchmark datasets in the field of person re-recognition. It has been widely used in the evaluation of algorithms, among others.

DukeMTMC-reID is a dataset released by the Duke Lab. The dataset contains over 1400 identified persons, captured from 2014 to 2015 by eight high-definition cameras at different locations on campus, with between 8 and 20 images per person, as well as additional information such as camera IDs, timestamps, and person identification tags. The complexity of the DukeMTMC-ReID dataset is associated with more image variations due to different viewing angles, lighting, and occlusion.

MSMT17 was presented by a research team from Peking University at the CVPR2018 conference, was collected from a network of 15 cameras on campus, including 12 outdoor cameras and 3 indoor cameras, and contained about 126,000 pedestrian images, covering 4101 different pedestrian identities. The data covers both indoor and outdoor scenes, and is collected in different time periods, with significant lighting changes, compared with other person re-identification datasets, MSMT17 is larger in terms of the number of pedestrians and the number of images.

It is important to note that the evaluation protocol for the three datasets is the same. In our experiments, we validate the performance of the model using the cumulative matching characteristics[41] (CMC) metrics, which include Rank-1, Rank-5, and Rank-10 precision, and the mean average precision (mAP) metric.

Implementation details

The method proposed in this paper uses ResNet-50[34] pre-trained on ImageNet[9] as the backbone model, and the hyperparameter tuning and specific configurations is the same as SPCL[26]. After layer 4, all the sub-module layers are removed, and the CAPF module is added, generating 2048 dimensional features. We use the DBSCAN algorithm as the clustering method. The size of the input images is 256×128, and each batch consists of 256 images, including 16 pseudo-labels, each containing 16 images. Random horizontal flipping, random cropping, and random erasure are performed for training images. The model was trained using an Adam optimizer with a weight decay of 5e−4. The number of clustering iterations was 50, and the learning rate was initialized to 3.5e−4 with a 90% decrease every 20 epochs. Clustering of k neighboring points is performed by Jaccard distance[42], and the value of k is set to 30. In the DBSCAN algorithm, the maximum distance d of two samples from each other is set to be 0.6, while ensuring that the minimum number of neighbors of a core point is 4.

Comparisons with existing unsupervised methods

Compared with SOTA USL methods being compared

To validate the effectiveness of our proposed method, we conducted comprehensive comparative experiments on three benchmark datasets: Market1501[38], DukeMTMC-reID[39], and MSMT17[40]. As shown in Table 1, the experimental results demonstrate that our method achieves superior performance compared to existing unsupervised methods, thereby confirming its effectiveness. Specifically, our approach shows significant

Method		Market-1501				DukeMTMC-reID				MSMT17			
		mAP	top-1	top-5	top-10	mAP	top-1	top-5	top-10	mAP	top-1	top-5	top-10
BUC[15]	AAAI'19	38.3	66.2	79.6	84.5	27.5	47.4	62.6	68.4	–	–	–	–
SSL[43]	CVPR'20	37.8	71.7	83.8	87.4	28.6	52.5	63.5	68.9	–	–	–	–
MMCL[19]	CVPR'20	45.5	80.3	89.4	92.3	51.4	72.4	82.9	85.0	15.1	40.8	51.8	56.7
HCT[3]	CVPR'20	56.4	80.0	91.6	95.2	50.7	69.6	83.4	87.4	–	–	–	–
CycAs[44]	ECCV'20	64.8	84.8	–	–	60.1	77.9	–	–	26.7	50.1	–	–
UGA[45]	ECCV'20	70.3	87.2	–	–	53.3	75.0	–	–	21.7	49.5	–	–
SPCL[26]	NeurIPS'20	73.1	88.1	95.1	97.0	65.3	81.2	90.3	92.2	22.7	47.7	60.0	65.5
CAP[46]	AAAI'21	79.2	91.4	96.3	97.7	67.3	81.1	89.3	91.8	36.9	67.4	78.8	81.4
ICE[20]	ICCV'21	82.3	93.8	97.6	98.4	69.9	83.3	91.5	94.1	38.9	70.2	80.5	84.4
IIDS[47]	TPAMI'22	78.0	91.2	96.2	97.7	68.7	82.1	90.8	93.7	26.9	56.4	68.8	73.4
SECRET[48]	AAAI'21	81.0	92.6	–	–	63.9	77.9	–	–	31.7	60.0	–	–
PPLR[21]	CVPR'22	81.5	92.8	97.1	98.1	–	–	–	–	42.2	73.3	83.5	86.5
AHCL	**Ours**	**84.5**	**94.6**	**97.8**	**98.6**	**72.1**	**85.3**	**92.7**	**95.1**	**45.1**	**75.9**	**86.8**	**89.1**

Table 1. Comparison with the state-of-the-art methods on Market1501, DukeMTMC-reID and MSMT17. Significant values are in bold.

Method	BUC[15]	SSL[43]	MMCL[19]	HCT[3]	CycAs[44]	UGA[45]
P value	1.3×10^{-4}	3.9×10^{-4}	9.3×10^{-5}	2.7×10^{-3}	2.8×10^{-3}	2.7×10^{-3}
Method	SPCL[26]	CAP[46]	ICE[20]	IIDS[47]	SECRET[48]	PPLR[21]
P value	2.7×10^{-3}	4.8×10^{-5}	1.9×10^{-3}	4.4×10^{-3}	1.3×10^{-2}	6.8×10^{-4}

Table 2. The results of paired t-test.

Method	Market-1501			
	mAP	Rank-1	Rank-5	Rank-10
ResNet50	84.5	94.6	97.8	98.6
IBN-ResNet50	86.4	95.5	98.1	98.8
IBN-ResNet50	46.2	77.6	88.3.2	90.3
Method	**DukeMTMC-reID**			
	mAP	**Rank-1**	**Rank-5**	**Rank-10**
ResNet50	72.1	85.3	92.7	95.1
IBN-ResNet50	74.5	86.9	93.6	95.7
Method	**MSMT17**			
	mAP	**Rank-1**	**Rank-5**	**Rank-10**
ResNet50	45.1	75.9	86.8	89.1
IBN-ResNet50	46.2	77.6	88.3.2	90.3

Table 3. Compare with other backbones on Market1501, DukeMTMC and MSMT17.

Components			No	Market-1501		DukeMTMC-reID	
CAPF	L_{ccl}	L_{icl}		mAP	Rank-1	Rank-5	Rank-10
			1	78.9	90.5	69.6	82.4
✓			2	82.5	92.9	71.1	83.7
	✓		3	80.4	91.3	70.4	83.0
		✓	4	79.7	90.9	70.1	82.8
✓	✓		5	83.9	94.1	71.8	84.9
✓		✓	6	82.8	93.3	71.3	84.2
	✓	✓	7	81.7	92.1	70.9	83.5
✓	✓	✓	8	**84.5**	**94.6**	**72.1**	**85.3**

Table 4. Ablation study on Market1501 and DukeMTMC-reID:The effectiveness of each components in our proposed method. AHCL includes CAPF model, the cluster contrast loss L_{ccl} instance contrast loss L_{icl}. Significant values are in bold.

improvements over the state-of-the-art unsupervised method PPLR[28], with increases of 3.0% in mAP and 1.8% in Rank-1 accuracy on the Market-1501. Furthermore, on the MSMT17, our method outperforms PPLR[28] by 3.1% in mAP and 2.6% in Rank-1 accuracy, respectively.

We employed a paired *t* test for statistical analysis of the experimental data to evaluate the effectiveness of the model. The paired t-test is suitable for comparing the performance of the same set of samples under different conditions, effectively controlling for the impact of individual differences on the results. We conducted statistical analysis on twelve sets of experimental results and recorded the *P* value as the evaluation metric. As shown in Table 2, since the *p* value is less than the significance level of 0.05, the experimental results are statistically significant. This finding further supports the superiority of our proposed model in pedestrian re-identification.

Compared with IBN-ResNet50 and ResNet50 backbones

Instance-batch normalization (IBN)[49] combines the adventages of instance normalization (IN)[50] in learning appearance invariance features and batch normalization (BN)[34] in learning content-related information. It has been demonstrated to be effective in supervised[15] object re-ID tasks. As Table 3 indicates, utilizing IBN-ResNet as the backbone can improve performance further.

Ablation studies

In this section, to validate the effectiveness of the various components of our proposed methodology, we conduct several detailed comparative experiments on two datasets, Market1501 and DukeMTMC. USL pipeline method

Components		Market-1501		DukeMTMC-reID	
Spatial attention	SENet	mAP	Rank-1	Rank-5	Rank-10
		78.9	90.0	69.6	82.4
✓		82.8	93.3	71.3	84.2
	✓	81.7	92.1	70.9	83.5
✓	✓	**84.5**	**94.6**	**72.1**	**85.3**

Table 5. An ablation study isolating the spatial and channel attention mechanisms on Market1501 and DukeMTMC-reID. Significant values are in bold.

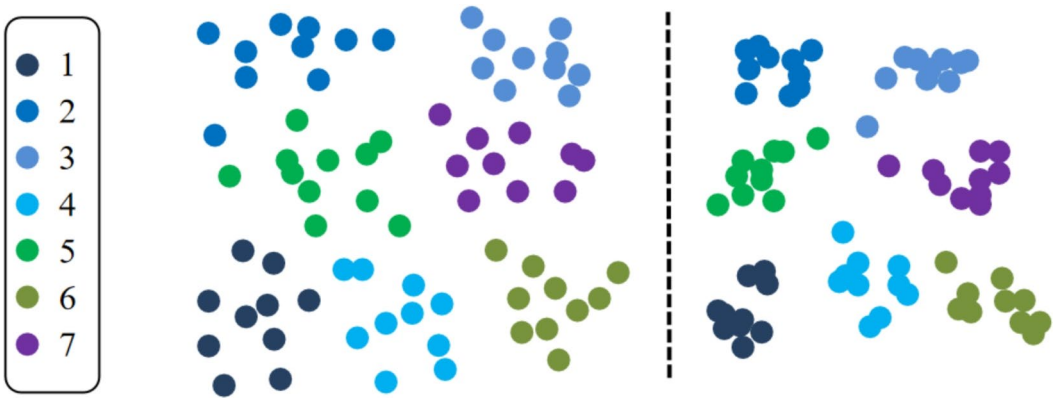


Fig. 5. Visualizations of feature distributions before and after applying the cluster-level contrastive loss.

with instance-level memory dictionaries is defined as our baseline methods. To verify the effectiveness of our three proposed components: the channel attention-based pooling feature fusion model(CAPF), the cluster contrast loss L_{ccl} and the instance contrast loss L_{icl}, we display the results in Table 4.

Effectiveness of the channel attention based pooling feature fusion model (CAPF)
The effectiveness of our proposed CAPF module can be seen through the following comparative experiments in (No.1 and No.2), (No.3 and No.5), (No.4 and No.6), and (No.7 and No.8) presented in Table 5. Regarding performance, CAPF significantly improved the baseline by 3.6%/2.4% and 1.5%/1.3% mAP/Rank-1 on Market1501 and DukeMTMC, respectively. By enhancing the backbone's feature extraction ability, CAPF made the feature extractor pay more attention to the differences between different samples and the similarities between the same samples, resulting in more representative extracted features. The experimental results show that CAPF is beneficial in allowing the model to extract more representative feature vectors.

An ablation study isolating the spatial and channel attention mechanisms would provide deeper insights into their individual contributions to the overall performance of the CAPF module, therefore, we have conducted additional experiments in which we separately evaluate the impact of the spatial attention mechanism and the channel attention mechanism. The experimental results in Table 5 show that the combined use of both mechanisms in the CAPF module leads to a synergistic effect, resulting in superior performance compared to using either mechanism alone.

Effectiveness of the cluster contrast loss L_{ccl}
The effectiveness of our proposed L_{ccl} can be seen through the following comparative experiments in (No.1 and No.3), (No.2 and No.5), (No.4 and No.7), and (No.6 and No.8) presented in Table 3. Suppose only one feature is selected in each cluster. In that case, it is difficult to represent the overall level of the whole cluster, which is not conducive to the network obtaining more discriminative information. L_{ccl} addressed this issue by enhancing the cluster-level discrimination and intra-cluster diversity, which resulted in significant performance improvements.

To validate the effectiveness of the proposed cluster-level contrastive loss, we performed a feature distribution visualization analysis and the results are illustrated in Fig. 5. Without the cluster-level contrastive loss, the feature distribution of different categories exhibits fuzzy inter class boundaries and large inter class spacing, indicating that the model has limited ability to distinguish samples from different categories, indicating the model's limited ability to distinguish samples from different categories. In contrast, after applying the cluster-level contrastive loss, the feature distributions become more compact within each category, and the separation between categories is significantly improved. These results demonstrate that the proposed loss function effectively enhances feature discriminability, facilitating intra-class compactness and inter-class separability.

	Market1501			
a	mAP	Rank-1	Rank-5	Rank-10
0	83.9	94.1	96.2	97.3
0.25	84.2	94.3	96.9	97.9
0.5	**84.5**	**94.6**	**97.8**	**98.6**
0.75	84.1	94.2	97.3	98.0
1	82.8	93.8	97.0	97.4

Table 6. Ablation study on Market1501:The results of different weight coefficient values. Significant values are in bold.

	Ours	
Method	mAP	Rank-1
32	74.8	87.9
64	80.4	91.6
128	82.3	93.1
256	**84.5**	**94.6**

Table 7. The impact of batch size to baseline method and our method on Market1501 dataset. Significant values are in bold.

Effectiveness of the instance contrast loss L_{icl}
The effectiveness of our proposed L_{icl} can be seen through the following comparative experiments in (No.1 and No.4), (No.2 and No.6), (No.3 and No.7), and (No.5 and No.8), as shown in Table 3. Using only instance-level samples for comparison is not conducive to learning generalized features and obtaining better clustering pseudo-labels. Therefore, it is observed that only when the cluster contrast loss and the instance contrast loss are used together can the module achieve its best performance.

To evaluate the robustness of the proposed mixing losses, it is necessary to introduce and analyze the weighting coefficients. Table 6 shows the experimental results of different weight coefficient values, and the experimental results show that the best effect is achieved when a is 0.5.

Batch size
Using a larger batch size during training can improve the stability of the training process and lead to better generalization performance. However, increasing the batch size requires more GPU memory and may result in slower training speed. The experimental results in Table 7 indicate that the performance of the proposed method remains stable over a wide range of batch sizes from 64 to 256, and using a larger batch size can achieve better results. Therefore, it is recommended to use a batch size of at least 64 to train the proposed method.

Conclusion and discussion

In this paper, we introduce a novel Attention-based Hybrid Contrastive Learning (AHCL) method for unsupervised person re-identification, which integrates an attention mechanism with a hybrid contrastive learning framework to enhance feature extraction and discrimination. The attention mechanism combines global average pooling and global max pooling to improve feature sensitivity, while the hybrid contrastive learning framework extracts both cluster-level and instance-level features from a dual memory dictionary, further boosting model performance. Experimental results demonstrate that AHCL outperforms compared methods on specific datasets and settings.

The AHCL method exhibits broad applicability in anomaly detection and general feature learning due to its core mechanisms: the attention mechanism enhances feature sensitivity, contrastive learning improves feature discrimination, pseudo-label generation mitigates label scarcity, and the hybrid framework enhances feature generality and robustness. However, several limitations remain, including challenges in negative sample selection, generalization of the attention mechanism, pseudo-label quality, and model scalability. Future research should address these issues to improve generalization and practical utility.

A critical aspect of AHCL is the computational complexity and scalability of its dual memory dictionaries. While this approach enhances instance-level and cluster-level feature discrimination, it introduces significant computational and memory demands. The complexity primarily stems from similarity calculations between query features and dictionary entries, with a computational cost of $O(N \times D)$, where N is the number of instances and D is the feature dimensionality. The InfoNCE loss function and clustering operations (e.g., DBSCAN) further increase the computational burden, with clustering complexity ranging from $O(N \log N)$ to $O(N^2)$. To address scalability challenges, strategies such as distributed storage, feature compression, optimized sampling, and approximate nearest neighbor clustering are proposed. These optimizations aim to balance performance and scalability, making AHCL more suitable for large-scale person re-identification tasks.

Data availability

The datasets used and analyzed during the current study are available from the corresponding author on reasonable request.

Received: 19 December 2024; Accepted: 7 April 2025
Published online: 17 April 2025

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Acknowledgements

This work was supported in part by the Chinese Natural Science Foundation (CNSF) (under Grant 61472278, Grant 61702165). This work was supported in part by the Major Project of Tianjin (under Grant 18ZXZNGX00150). This work was supported in part by the National Social Science Funds of China (NSSFC) (under Grant 23BKS169). This work was supported in part by the Hebei Provincial Natural Science Foundation, China (under Grant No. F2020111001). This work was supported in part by the Scientific Research Project of Hengshui University (under Grant 2023XJZX19, Grant 2023XJZX01, Grant 2023XTCX026).

Author contributions

Conceptualization, W.Q. and Y.L.; methodology, W.Q.; software, W.Q. and Q.G.; validation, J.Z.; writing—original draft preparation, J.Z. and Y.L.; writing—review and editing, X.W. and J.G. All authors have read and agreed to the published version of the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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